

Week 1 Report

Progress this week:

- Completed initial onboarding for the photogrammetry project
- Reviewed the project overview
- Studied structure of pipeline:
 - Image intake
 - Masking
 - COLMAP pose estimation
 - Gaussian reconstruction
 - Mesh extraction
- Reviewed previous summaries and results from previous work
- Familiarized myself with the reconstruction methods currently being considered like 2DGS, PGSR, SuGaR and Gaussian Wrapping
- Reviewed the proposed dataset/benchmark paper direction

Next Week

- Continue looking through codebase
- Confirm data and compute access
- Investigate the two proposed starter tasks + colour calibration
- Continue reading relevant papers

Abstracts

Structure-from-Motion Revisited

- Incremental Structure-from-Motion is a prevalent strategy for 3D reconstruction from unordered image collections. While incremental reconstruction systems have tremendously advanced in all regards, robustness, accuracy, completeness, and scalability remain the key problems towards building a truly general-purpose pipeline. We propose a new SfM technique that improves upon the state of the art to make a further step towards this ultimate goal. The full reconstruction pipeline is released to the public as an open-source implementation.

2D Gaussian Splatting for Geometrically Accurate Radiance Fields

- 3D Gaussian Splatting (3DGS) has recently revolutionized radiance field reconstruction, achieving high quality novel view synthesis and fast rendering speed without baking.

However, 3DGS fails to accurately represent surfaces due to the multi-view inconsistent nature of 3D Gaussians. We present 2D Gaussian Splatting (2DGS), a novel approach to model and reconstruct geometrically accurate radiance fields from multi-view images. Our key idea is to collapse the 3D volume into a set of 2D oriented planar Gaussian disks. Unlike 3D Gaussians, 2D Gaussians provide view-consistent geometry while modeling surfaces intrinsically. To accurately recover thin surfaces and achieve stable optimization, we introduce a perspective-correct 2D splatting process utilizing ray-splat intersection and rasterization. Additionally, we incorporate depth distortion and normal consistency terms to further enhance the quality of the reconstructions. We demonstrate that our differentiable renderer allows for noise-free and detailed geometry reconstruction while maintaining competitive appearance quality, fast training speed, and real-time rendering.